The leading constant for $S_n \times A$

Let $n \in \{3,4,5\}$, and let A be a finite abelian group of order $m = |A|$. Then we consider $G = S_n \times A$. Write $N_G(X)$ for the number of $\mathbb{Q}$-isomorphism classes of degree mn fields L with $\mathrm{Gal}(L^c/\mathbb{Q}) \simeq G$ and $|\operatorname{Disc} L| < X$. All signatures are included. The constant in $N_G(X) \sim C_G X^{1/m}$ as $X \to \infty$ is given by

$$C_G := \frac{\kappa_n}{|\operatorname{Aut} A|} \sum_{B \le A} \mu(B,A) \prod_{p \text{ prime}} \left[(1-p^{-1}) \sum_{\substack{\psi \in \mathcal{H}_p(B) \\ E \in \mathcal{E}_n(p)}} \frac{p^{-d_p(E \otimes_{\mathbb{Q}_p} L_\psi)/h_\psi}}{|\operatorname{Aut}_{\mathbb{Q}_p}(E)|} \right].$$

Notation. The constants are $\kappa_3 = 1/3$, $\kappa_4 = 5/24$ and $\kappa_5 = 13/120$. The sum is over all subgroups B of A. The Möbius function is defined by

$$\mu(B,A) := \begin{cases} \displaystyle\prod_{p \text{ prime}} (-1)^{r_p} p^{\frac{r_p(r_p-1)}{2}} & \text{if } A/B \cong \prod_p (\mathbb{Z}/p\mathbb{Z})^{r_p}, \\ 0, & \text{otherwise.} \end{cases}$$

For a prime p, the finite set $\mathcal{H}_p(B)$ consists of all continuous group homomorphisms $\psi : \mathbb{Z}_p^\times \to B$. Extend ψ to $\mathbb{Q}_p^\times$ by setting $\psi(p) = 1$, where 1 is the identity of B. Local class field theory assigns to the kernel of this extension a finite abelian extension $L_\psi/\mathbb{Q}_p$ of degree $h_\psi = |\operatorname{im} \psi|$.

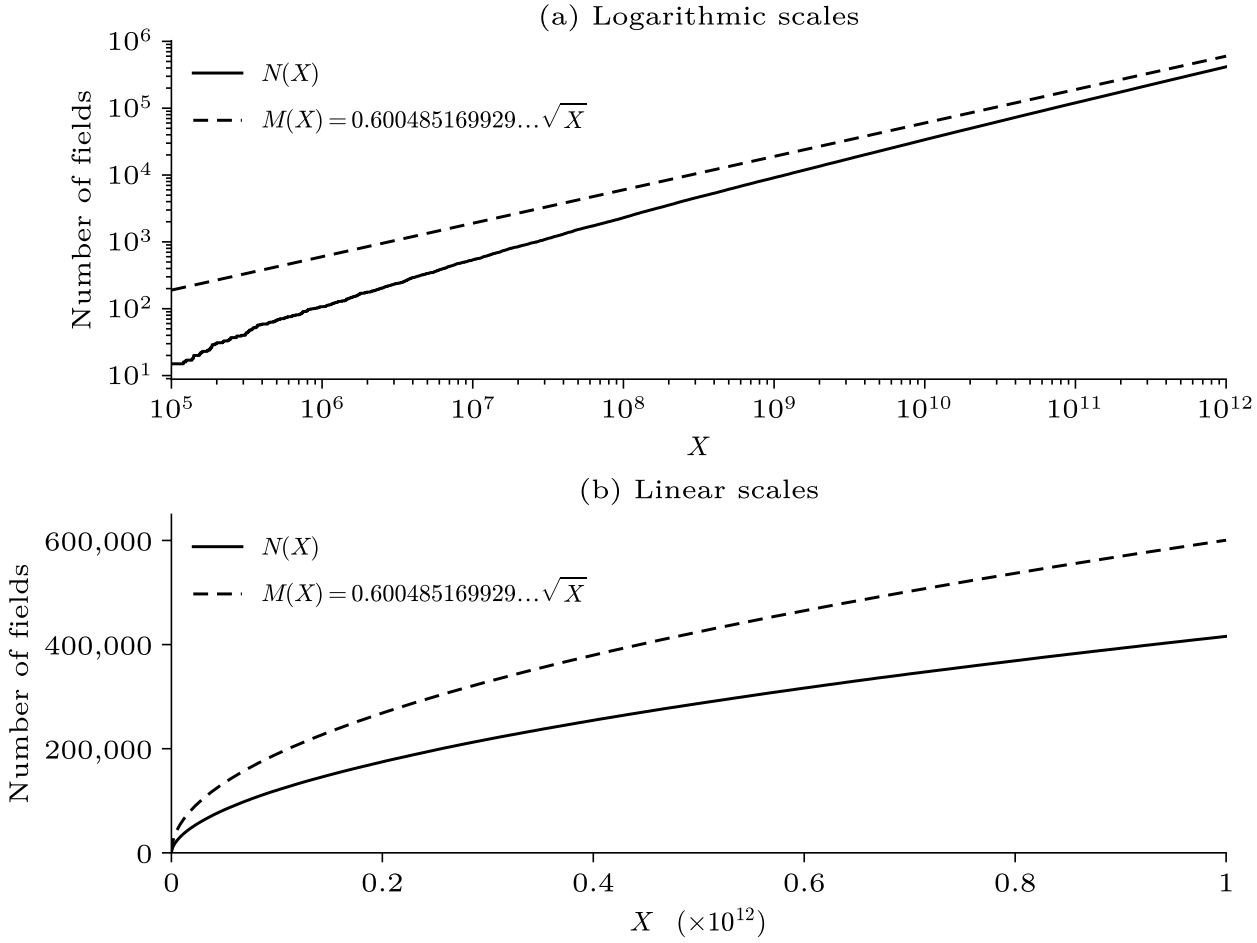
Let $\mathcal{E}_n(p)$ be the finite set of isomorphism classes of degree n étale $\mathbb{Q}_p$-algebras. For $E \in \mathcal{E}_n(p)$, let $d_p(E \otimes_{\mathbb{Q}_p} L_\psi)$ be the discriminant exponent of the maximal $\mathbb{Z}_p$-order in $E \otimes_{\mathbb{Q}_p} L_\psi$. Finally, $\operatorname{Aut}_{\mathbb{Q}_p}(E)$ is the group of $\mathbb{Q}_p$-algebra automorphisms of E.

Some numerics

n	$C_{S_n \times C_2}$	$C_{S_n \times (C_2 \times C_2)}$
3	0.600485169929	0.685009888568
4	0.274102386235	0.131474579489
5	0.067505846256	0.012294591845

The asymptotic powers are $X^{1/2}$ and $X^{1/4}$ respectively.

For $G = S_3 \times C_2$, the two functions below are the actual count $N(X) = N_{S_3\times C_2}(X)$ and its leading term $M(X) = C_{S_3\times C_2}\sqrt{X} = (0.600485169929\ldots)\sqrt{X}$. The solid curve represents $N(X)$ and the dashed curve represents $M(X)$.



Some numerics

X	$N(X)$	$M(X)$
10^6	108	600.49
10^8	2 299	6 004.85
10^{10}	33 770	60 048.52
10^{11}	120 217	189 890.08
10^{12}	415 740	600 485.17

The counts include all signatures and are complete for $X \le 10^{12}$ by using the PARI/GP cubic tables and all relevant quadratic fields. Their discriminant and signature multiplicities below 10^7 agree with the LMFDB.